

OOP

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Chapter 1

Objektinis Programavimas

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

Zmogus	21
Student	9

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Student		9
Zmogus		21

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

deque_test.cpp	23
func_masyvas.h	24
functions.cpp	26
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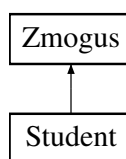
Chapter 5

Class Documentation

5.1 Student Class Reference

```
#include <func_masyvas.h>
```

Inheritance diagram for Student:



Public Member Functions

- [Student](#) ()
- [Student](#) (const std::string &v, const std::string &p, const std::vector< int > &nd, int e)
- void [setVardas](#) (const std::string &v)
- void [setPavarde](#) (const std::string &p)
- void [setNamuDarbai](#) (const std::vector< int > &nd)
- void [setEgz](#) (int e)
- std::string [getVardas](#) () const
- std::string [getPavarde](#) () const
- std::vector< int > [getNamuDarbai](#) () const
- int [getEgz](#) () const
- double [getGalutinisVid](#) () const
- double [getGalutinisMed](#) () const
- double [skaiciuotiVidurki](#) () const
- double [skaiciuotiMediana](#) () const
- double [skaiciuotiGalutiniVidurki](#) () const
- double [skaiciuotiGalutiniMediana](#) () const
- [Student](#) ()
- [Student](#) (const std::string &v, const std::string &p, const std::vector< int > &nd, int e)
- void [setVardas](#) (const std::string &v)
- void [setPavarde](#) (const std::string &p)
- void [setNamuDarbai](#) (const std::vector< int > &nd)
- void [setEgz](#) (int e)

- `std::string getVardas () const`
- `std::string getPavarde () const`
- `std::vector< int > getNamuDarbai () const`
- `int getEgz () const`
- `double getGalutinisVid () const`
- `double getGalutinisMed () const`
- `double skaiciuotiVidurki () const`
- `double skaiciuotiMediana () const`
- `double skaiciuotiGalutiniVidurki () const`
- `double skaiciuotiGalutiniMediana () const`
- `Student ()`
- `Student (const std::string &v, const std::string &p, const std::vector< int > &nd, int e)`
- `Student (const Student &old_stud)`
- `Student (Student &&other) noexcept`
- `Student & operator= (const Student &other)`
- `Student & operator= (Student &&other) noexcept`
- `~Student ()=default`
- `void setVardas (const std::string &v)`
- `void setPavarde (const std::string &p)`
- `void setNamuDarbai (const std::vector< int > &nd)`
- `void setEgz (int e)`
- `std::string getVardas () const`
- `std::string getPavarde () const`
- `std::vector< int > getNamuDarbai () const`
- `int getEgz () const`
- `double getGalutinisVid () const`
- `double getGalutinisMed () const`
- `double skaiciuotiVidurki () const`
- `double skaiciuotiMediana () const`
- `double skaiciuotiGalutiniVidurki () const`
- `double skaiciuotiGalutiniMediana () const`
- `Student ()`
- `Student (const std::string &v, const std::string &p, const std::vector< int > &nd, int e)`
- `void setVardas (const std::string &v)`
- `void setPavarde (const std::string &p)`
- `void setNamuDarbai (const std::vector< int > &nd)`
- `void setEgz (int e)`
- `std::string getVardas () const`
- `std::string getPavarde () const`
- `std::vector< int > getNamuDarbai () const`
- `int getEgz () const`
- `double getGalutinisVid () const`
- `double getGalutinisMed () const`
- `double skaiciuotiVidurki () const`
- `double skaiciuotiMediana () const`
- `double skaiciuotiGalutiniVidurki () const`
- `double skaiciuotiGalutiniMediana () const`

Public Member Functions inherited from **Zmogus**

- `Zmogus ()=default`
- `Zmogus (const std::string &v, const std::string &p)`
- `virtual ~Zmogus ()=default`

Static Public Member Functions

- static double [Vidurkis](#) (const std::vector< int > &pazymiai)
- static double [Vidurkis](#) (const std::vector< int > &pazymiai)
- static double [Vidurkis](#) (const std::vector< int > &pazymiai)
- static double [Vidurkis](#) (const std::vector< int > &pazymiai)

Public Attributes

- int * [namuDarbai](#)
- int [nd_kiek](#)
- std::string [name](#)
- double [grade](#)

Private Attributes

- std::string [vardas](#)
- std::string [pavarde](#)
- std::vector< int > [namuDarbai](#)
- int [egz](#)

Friends

- std::ostream & [operator<<](#) (std::ostream &os, const [Student](#) &s)
- std::istream & [operator>>](#) (std::istream &is, [Student](#) &s)

Additional Inherited Members

Protected Attributes inherited from [Zmogus](#)

- std::string [vardas](#)
- std::string [pavarde](#)

5.1.1 Constructor & Destructor Documentation

5.1.1.1 [Student\(\)](#) [1/10]

```
Student::Student () [inline]
```

5.1.1.2 [Student\(\)](#) [2/10]

```
Student::Student (
    const std::string & v,
    const std::string & p,
    const std::vector< int > & nd,
    int e) [inline]
```

5.1.1.3 Student() [3/10]

```
Student::Student () [inline]
```

5.1.1.4 Student() [4/10]

```
Student::Student (  
    const std::string & v,  
    const std::string & p,  
    const std::vector< int > & nd,  
    int e) [inline]
```

5.1.1.5 Student() [5/10]

```
Student::Student () [inline]
```

5.1.1.6 Student() [6/10]

```
Student::Student (  
    const std::string & v,  
    const std::string & p,  
    const std::vector< int > & nd,  
    int e) [inline]
```

5.1.1.7 Student() [7/10]

```
Student::Student (  
    const Student & old_stud) [inline]
```

5.1.1.8 Student() [8/10]

```
Student::Student (  
    Student && other) [inline], [noexcept]
```

5.1.1.9 ~Student()

```
Student::~~Student () [default]
```

5.1.1.10 Student() [9/10]

```
Student::Student () [inline]
```

5.1.1.11 Student() [10/10]

```
Student::Student (
    const std::string & v,
    const std::string & p,
    const std::vector< int > & nd,
    int e) [inline]
```

5.1.2 Member Function Documentation

5.1.2.1 getEgz() [1/4]

```
int Student::getEgz () const [inline]
```

5.1.2.2 getEgz() [2/4]

```
int Student::getEgz () const [inline]
```

5.1.2.3 getEgz() [3/4]

```
int Student::getEgz () const [inline]
```

5.1.2.4 getEgz() [4/4]

```
int Student::getEgz () const [inline]
```

5.1.2.5 getGalutinisMed() [1/4]

```
double Student::getGalutinisMed () const [inline]
```

5.1.2.6 getGalutinisMed() [2/4]

```
double Student::getGalutinisMed () const [inline]
```

5.1.2.7 getGalutinisMed() [3/4]

```
double Student::getGalutinisMed () const [inline]
```

5.1.2.8 getGalutinisMed() [4/4]

```
double Student::getGalutinisMed () const [inline]
```

5.1.2.9 getGalutinisVid() [1/4]

```
double Student::getGalutinisVid () const [inline]
```

5.1.2.10 getGalutinisVid() [2/4]

```
double Student::getGalutinisVid () const [inline]
```

5.1.2.11 getGalutinisVid() [3/4]

```
double Student::getGalutinisVid () const [inline]
```

5.1.2.12 getGalutinisVid() [4/4]

```
double Student::getGalutinisVid () const [inline]
```

5.1.2.13 getNamuDarbai() [1/4]

```
std::vector< int > Student::getNamuDarbai () const [inline]
```

5.1.2.14 getNamuDarbai() [2/4]

```
std::vector< int > Student::getNamuDarbai () const [inline]
```

5.1.2.15 getNamuDarbai() [3/4]

```
std::vector< int > Student::getNamuDarbai () const [inline]
```

5.1.2.16 getNamuDarbai() [4/4]

```
std::vector< int > Student::getNamuDarbai () const [inline]
```

5.1.2.17 getPavarde() [1/4]

```
std::string Student::getPavarde () const [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.18 getPavarde() [2/4]

```
std::string Student::getPavarde () const [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.19 getPavarde() [3/4]

```
std::string Student::getPavarde () const [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.20 getPavarde() [4/4]

```
std::string Student::getPavarde () const [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.21 getVardas() [1/4]

```
std::string Student::getVardas () const [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.22 getVardas() [2/4]

```
std::string Student::getVardas () const [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.23 getVardas() [3/4]

```
std::string Student::getVardas () const [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.24 getVardas() [4/4]

```
std::string Student::getVardas () const [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.25 operator=() [1/2]

```
Student & Student::operator= (  
    const Student & other) [inline]
```

5.1.2.26 operator=() [2/2]

```
Student & Student::operator= (  
    Student && other) [inline], [noexcept]
```

5.1.2.27 setEgz() [1/4]

```
void Student::setEgz (
    int e) [inline]
```

5.1.2.28 setEgz() [2/4]

```
void Student::setEgz (
    int e) [inline]
```

5.1.2.29 setEgz() [3/4]

```
void Student::setEgz (
    int e) [inline]
```

5.1.2.30 setEgz() [4/4]

```
void Student::setEgz (
    int e) [inline]
```

5.1.2.31 setNamuDarbai() [1/4]

```
void Student::setNamuDarbai (
    const std::vector< int > & nd) [inline]
```

5.1.2.32 setNamuDarbai() [2/4]

```
void Student::setNamuDarbai (
    const std::vector< int > & nd) [inline]
```

5.1.2.33 setNamuDarbai() [3/4]

```
void Student::setNamuDarbai (
    const std::vector< int > & nd) [inline]
```

5.1.2.34 setNamuDarbai() [4/4]

```
void Student::setNamuDarbai (
    const std::vector< int > & nd) [inline]
```

5.1.2.35 setPavarde() [1/4]

```
void Student::setPavarde (
    const std::string & p) [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.36 setPavarde() [2/4]

```
void Student::setPavarde (  
    const std::string & p) [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.37 setPavarde() [3/4]

```
void Student::setPavarde (  
    const std::string & p) [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.38 setPavarde() [4/4]

```
void Student::setPavarde (  
    const std::string & p) [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.39 setVardas() [1/4]

```
void Student::setVardas (  
    const std::string & v) [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.40 setVardas() [2/4]

```
void Student::setVardas (  
    const std::string & v) [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.41 setVardas() [3/4]

```
void Student::setVardas (  
    const std::string & v) [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.42 setVardas() [4/4]

```
void Student::setVardas (  
    const std::string & v) [inline], [virtual]
```

Implements [Zmogus](#).

5.1.2.43 skaiciuotiGalutiniMediana() [1/4]

```
double Student::skaiciuotiGalutiniMediana () const [inline]
```

5.1.2.44 skaiciuotiGalutiniMediana() [2/4]

```
double Student::skaiciuotiGalutiniMediana () const [inline]
```

5.1.2.45 skaiciuotiGalutiniMediana() [3/4]

```
double Student::skaiciuotiGalutiniMediana () const [inline]
```

5.1.2.46 skaiciuotiGalutiniMediana() [4/4]

```
double Student::skaiciuotiGalutiniMediana () const [inline]
```

5.1.2.47 skaiciuotiGalutiniVidurki() [1/4]

```
double Student::skaiciuotiGalutiniVidurki () const [inline]
```

5.1.2.48 skaiciuotiGalutiniVidurki() [2/4]

```
double Student::skaiciuotiGalutiniVidurki () const [inline]
```

5.1.2.49 skaiciuotiGalutiniVidurki() [3/4]

```
double Student::skaiciuotiGalutiniVidurki () const [inline]
```

5.1.2.50 skaiciuotiGalutiniVidurki() [4/4]

```
double Student::skaiciuotiGalutiniVidurki () const [inline]
```

5.1.2.51 skaiciuotiMediana() [1/4]

```
double Student::skaiciuotiMediana () const [inline]
```

5.1.2.52 skaiciuotiMediana() [2/4]

```
double Student::skaiciuotiMediana () const [inline]
```

5.1.2.53 skaiciuotiMediana() [3/4]

```
double Student::skaiciuotiMediana () const [inline]
```

5.1.2.54 skaiciuotiMediana() [4/4]

```
double Student::skaiciuotiMediana () const [inline]
```

5.1.2.55 skaiciuotiVidurki() [1/4]

```
double Student::skaiciuotiVidurki () const [inline]
```

5.1.2.56 skaiciuotiVidurki() [2/4]

```
double Student::skaiciuotiVidurki () const [inline]
```

5.1.2.57 skaiciuotiVidurki() [3/4]

```
double Student::skaiciuotiVidurki () const [inline]
```

5.1.2.58 skaiciuotiVidurki() [4/4]

```
double Student::skaiciuotiVidurki () const [inline]
```

5.1.2.59 Vidurkis() [1/4]

```
static double Student::Vidurkis (  
    const std::vector< int > & pazymiai) [inline], [static]
```

5.1.2.60 Vidurkis() [2/4]

```
static double Student::Vidurkis (  
    const std::vector< int > & pazymiai) [inline], [static]
```

5.1.2.61 Vidurkis() [3/4]

```
static double Student::Vidurkis (  
    const std::vector< int > & pazymiai) [inline], [static]
```

5.1.2.62 Vidurkis() [4/4]

```
static double Student::Vidurkis (  
    const std::vector< int > & pazymiai) [inline], [static]
```

5.1.3 Friends And Related Symbol Documentation

5.1.3.1 operator<<

```
std::ostream & operator<< (  
    std::ostream & os,  
    const Student & s) [friend]
```

5.1.3.2 operator>>

```
std::istream & operator>> (  
    std::istream & is,  
    Student & s) [friend]
```

5.1.4 Member Data Documentation

5.1.4.1 egz

```
int Student::egz [private]
```

5.1.4.2 grade

```
double Student::grade
```

5.1.4.3 name

```
std::string Student::name
```

5.1.4.4 namuDarbai [1/2]

```
std::vector< int > Student::namuDarbai [private]
```

5.1.4.5 namuDarbai [2/2]

```
int* Student::namuDarbai
```

5.1.4.6 nd_kiek

```
int Student::nd_kiek
```

5.1.4.7 pavarde

```
std::string Student::pavarde [private]
```

5.1.4.8 vardas

```
std::string Student::vardas [private]
```

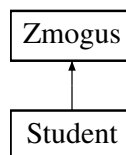
The documentation for this class was generated from the following files:

- [deque_test.cpp](#)
- [func_masyvas.h](#)
- [kont_test.cpp](#)
- [list_test.cpp](#)
- [student.h](#)
- [vector_test.cpp](#)

5.2 Zmogus Class Reference

```
#include <zmogus.h>
```

Inheritance diagram for Zmogus:



Public Member Functions

- [Zmogus](#) ()=default
- [Zmogus](#) (const std::string &v, const std::string &p)
- virtual [~Zmogus](#) ()=default
- virtual std::string [getVardas](#) () const =0
- virtual std::string [getPavarde](#) () const =0
- virtual void [setVardas](#) (const std::string &v)=0
- virtual void [setPavarde](#) (const std::string &p)=0

Protected Attributes

- std::string [vardas](#)
- std::string [pavarde](#)

5.2.1 Constructor & Destructor Documentation

5.2.1.1 Zmogus() [1/2]

```
Zmogus::Zmogus () [default]
```

5.2.1.2 Zmogus() [2/2]

```
Zmogus::Zmogus (
    const std::string & v,
    const std::string & p) [inline]
```

5.2.1.3 ~Zmogus()

```
virtual Zmogus::~Zmogus () [virtual], [default]
```

5.2 Member Function Documentation

5.2.2.1 getPavarde()

```
virtual std::string Zmogus::getPavarde () const [pure virtual]
```

Implemented in [Student](#), [Student](#), [Student](#), and [Student](#).

5.2.2.2 getVardas()

```
virtual std::string Zmogus::getVardas () const [pure virtual]
```

Implemented in [Student](#), [Student](#), [Student](#), and [Student](#).

5.2.2.3 setPavarde()

```
virtual void Zmogus::setPavarde (
    const std::string & p) [pure virtual]
```

Implemented in [Student](#), [Student](#), [Student](#), and [Student](#).

5.2.2.4 setVardas()

```
virtual void Zmogus::setVardas (
    const std::string & v) [pure virtual]
```

Implemented in [Student](#), [Student](#), [Student](#), and [Student](#).

5.2.3 Member Data Documentation

5.2.3.1 pavarde

```
std::string Zmogus::pavarde [protected]
```

5.2.3.2 vardas

```
std::string Zmogus::vardas [protected]
```

The documentation for this class was generated from the following file:

- [zmogus.h](#)

Chapter 6

File Documentation

6.1 deque_test.cpp File Reference

```
#include <iostream>
#include <fstream>
#include <vector>
#include <string>
#include <algorithm>
#include <numeric>
#include <iomanip>
#include <chrono>
#include <deque>
```

Classes

- class [Student](#)

Functions

- void [failo_nuskaitymas](#) (const std::string &pav, std::deque< [Student](#) > &studentai)
- void [padalinti1](#) (const std::deque< [Student](#) > &studentai, std::deque< [Student](#) > &kietekai, std::deque< [Student](#) > &vargsiukai)
- void [padalinti2](#) (std::deque< [Student](#) > &studentai, std::deque< [Student](#) > &vargsiukai)
- void [padalinti3](#) (std::deque< [Student](#) > &studentai, std::deque< [Student](#) > &vargsiukai)
- void [issaugoti_studentus](#) (const std::deque< [Student](#) > &studentai, const std::string &failo_pav)
- int [main](#) ()

6.1.1 Function Documentation

6.1.1.1 failo_nuskaitymas()

```
void failo_nuskaitymas (
    const std::string & pav,
    std::deque< Student > & studentai)
```

6.1.1.2 issaugoti_studentus()

```
void issaugoti_studentus (
    const std::deque< Student > & studentai,
    const std::string & failo_pav)
```

6.1.1.3 main()

```
int main ()
```

6.1.1.4 padalinti1()

```
void padalinti1 (
    const std::deque< Student > & studentai,
    std::deque< Student > & kietekai,
    std::deque< Student > & vargsiukai)
```

6.1.1.5 padalinti2()

```
void padalinti2 (
    std::deque< Student > & studentai,
    std::deque< Student > & vargsiukai)
```

6.1.1.6 padalinti3()

```
void padalinti3 (
    std::deque< Student > & studentai,
    std::deque< Student > & vargsiukai)
```

6.2 func_masyvas.h File Reference

```
#include <random>
#include <iostream>
#include <iomanip>
#include <vector>
#include <string>
#include <algorithm>
```

Classes

- class [Student](#)

Functions

- double [Vidurkis](#) (int *pazymiai)
- double [apskaiciuotiMediana](#) (int *pazymiai)
- void [Duom](#) ([Student](#) &studentas)
- void [Rez](#) (const [Student](#) &studentas)
- void [generuotiVardaPavarde](#) (std::string &vardas, std::string &pavarde)

6.2.1 Function Documentation

6.2.1.1 [apskaiciuotiMediana\(\)](#)

```
double apskaiciuotiMediana (  
    int * pazymiai)
```

6.2.1.2 [Duom\(\)](#)

```
void Duom (  
    Student & studentas)
```

6.2.1.3 [generuotiVardaPavarde\(\)](#)

```
void generuotiVardaPavarde (  
    std::string & vardas,  
    std::string & pavarde)
```

6.2.1.4 [Rez\(\)](#)

```
void Rez (  
    const Student & studentas)
```

6.2.1.5 [Vidurkis\(\)](#)

```
double Vidurkis (  
    int * pazymiai)
```

6.3 func_masyvas.h

[Go to the documentation of this file.](#)

```
00001 #ifndef FUNC_MASYVAS_H
00002 #define FUNC_MASYVAS_H
00003
00004 #include <random>
00005 #include <iostream>
00006 #include <iomanip>
00007 #include <vector>
00008 #include <string>
00009 #include <algorithm>
00010
00011 struct Student {
00012     std::string vardas;
00013     std::string pavarde;
00014     int* namuDarbai;
00015     int nd_kiek;
00016     int egz;
00017 };
00018
00019 double Vidurkis(int* pazymiai);
00020 double apskaiciuotiMediana(int* pazymiai);
00021 void Duom(Student& studentas);
00022 void Rez(const Student& studentas);
00023 void generuotiVardaPavarde(std::string& vardas, std::string& pavarde); // Nauja funkcija
00024
00025 #endif
```

6.4 functions.cpp File Reference

```
#include "functions.h"
#include <iostream>
#include <fstream>
#include <iomanip>
#include <algorithm>
#include <random>
#include <chrono>
```

Functions

- double [apskaiciuotiMediana](#) (std::vector< int > pazymiai)
- void [gen_name](#) (std::string &vardas, std::string &pavarde)
- void [gen_file](#) (const std::string &pav, int kiek)
- void [failo_nuskaitymas](#) (const std::string &pav, int kiek, std::vector< [Student](#) > &studentai)
- void [padalinti_studentus](#) (std::vector< [Student](#) > &studentai, std::vector< [Student](#) > &vargsiukai, std::vector< [Student](#) > &kietekai)
- void [issaugoti_studentus](#) (const std::vector< [Student](#) > &studentai, const std::string &failo_pav)
- void [Duom](#) (std::vector< [Student](#) > &studentai, int &ch1)
- void [Rez](#) (std::vector< [Student](#) > &studentai, int &ch1)

6.4.1 Function Documentation

6.4.1.1 apskaiciuotiMediana()

```
double apskaiciuotiMediana (
    std::vector< int > pazymiai)
```

6.4.1.2 Duom()

```
void Duom (
    std::vector< Student > & studentai,
    int & chl)
```

6.4.1.3 failo_nuskaitymas()

```
void failo_nuskaitymas (
    const std::string & pav,
    int kiek,
    std::vector< Student > & studentai)
```

6.4.1.4 gen_file()

```
void gen_file (
    const std::string & pav,
    int kiek)
```

6.4.1.5 gen_name()

```
void gen_name (
    std::string & vardas,
    std::string & pavarde)
```

6.4.1.6 issaugoti_studentus()

```
void issaugoti_studentus (
    const std::vector< Student > & studentai,
    const std::string & failo_pav)
```

6.4.1.7 padalinti_studentus()

```
void padalinti_studentus (
    std::vector< Student > & studentai,
    std::vector< Student > & vargsiukai,
    std::vector< Student > & kietekai)
```

6.4.1.8 Rez()

```
void Rez (
    std::vector< Student > & studentai,
    int & chl)
```

6.5 functions.cpp

[Go to the documentation of this file.](#)

```

00001 #include "functions.h"
00002 #include <iostream>
00003 #include <fstream>
00004 #include <iomanip>
00005 #include <algorithm>
00006 #include <random>
00007 #include <chrono>
00008
00009 double apskaiciuotiMediana(std::vector<int> pazymiai) {
00010     if (pazymiai.empty()) return 0.0;
00011     std::sort(pazymiai.begin(), pazymiai.end());
00012     size_t dydis = pazymiai.size();
00013     if (dydis % 2 == 0) {
00014         return (pazymiai[dydis / 2 - 1] + pazymiai[dydis / 2]) / 2.0;
00015     } else {
00016         return pazymiai[dydis / 2];
00017     }
00018 }
00019
00020 void gen_name(std::string& vardas, std::string& pavarde) {
00021     std::vector<std::string> vardai = {"Pijus", "Justas", "Kostas", "Petras"};
00022     std::vector<std::string> pavardes = {"Savanevicius", "Andru", "Baranauskas", "Dovydaitis"};
00023     std::random_device rd;
00024     std::mt19937 gen(rd());
00025     std::uniform_int_distribution<> distr(0, vardai.size() - 1);
00026     vardas = vardai[distr(gen)];
00027     pavarde = pavardes[distr(gen)];
00028 }
00029 void gen_file(const std::string& pav, int kiek)
00030 {
00031     auto start = std::chrono::high_resolution_clock::now();
00032     std::ofstream gf(pav);
00033     if (!gf) std::cout<<"failas nebuvo sukurtas"<<std::endl;
00034     std::random_device rd;
00035     std::mt19937 gen(rd());
00036     std::uniform_int_distribution<int> distr(1, 10);
00037     for (int i=0; i<kiek; i++)
00038     {
00039         gf<<"VardasNR"<i<" " <<"PavardeNR"<i;
00040         for (int j=0; j<5; j++)
00041         {
00042             gf<<" "<<distr(gen);
00043         }
00044         gf<<" ";
00045         gf<<distr(gen)<<std::endl;
00046     }
00047     auto end = std::chrono::high_resolution_clock::now();
00048     std::chrono::duration<double> diff = end - start;
00049     std::cout << "Failo " << pav << " generavimas užtruko: " << diff.count() << " s" << std::endl;
00050     gf.close();
00051 }
00052 void failo_nuskaitymas(const std::string& pav, int kiek, std::vector<Student>& studentai)
00053 {
00054     std::ifstream rf(pav);
00055     if (!rf) std::cout<<"failas neatsidaro"<<std::endl;
00056     std::string vardas, pavarde;
00057     while (rf >> vardas >> pavarde) {
00058         std::vector<int> namuDarbai;
00059         for (int i = 0; i < 5; i++) {
00060             int pazymys;
00061             rf >> pazymys;
00062             namuDarbai.push_back(pazymys);
00063         }
00064         int egz;
00065         rf >> egz;
00066         Student s;
00067         s.setVardas(vardas);
00068         s.setPavarde(pavarde);
00069         s.setNamuDarbai(namuDarbai);
00070         s.setEgz(egz);
00071         studentai.push_back(s);
00072     }
00073 }
00074 void padalinti_studentus (std::vector<Student>& studentai, std::vector<Student>& vargsiukai,
std::vector<Student>& kietekai)
00075 {
00076     auto start = std::chrono::high_resolution_clock::now();
00077     for (const auto& s : studentai)
00078     {
00079         if (s.getGalutinisVid() < 5.0)
00080         {
00081             vargsiukai.push_back(s);

```

```

00082     }
00083     else kietekai.push_back(s);
00084 }
00085 auto end = std::chrono::high_resolution_clock::now();
00086 std::chrono::duration<double> diff = end - start;
00087 std::cout << "Studentų padalijimas užtruko: " << diff.count() << " s" << std::endl;
00088 }
00089 void issaugoti_studentus(const std::vector<Student>& studentai, const std::string& failo_pav) {
00090     std::ofstream out(failo_pav);
00091     if (!out) {
00092         std::cerr << "Klaida atidarant failą: " << failo_pav << std::endl;
00093         return;
00094     }
00095
00096     out << std::fixed << std::setprecision(2);
00097     out << std::setw(15) << "Vardas" << std::setw(15) << "Pavarde" << std::setw(15) << "Galutinis (vid.)\n";
00098     out << "-----\n";
00099
00100     for (const auto& s : studentai) {
00101         out << std::setw(15) << s.getVardas()
00102             << std::setw(15) << s.getPavarde()
00103             << std::setw(15) << s.getGalutinisVid() << "\n";
00104     }
00105
00106     out.close();
00107 }
00108
00109 void Duom(std::vector<Student>& studentai, int& chl) {
00110     std::random_device rd;
00111     std::mt19937 gen(rd());
00112     std::uniform_int_distribution<> distr(1, 10);
00113
00114     std::string kitas_stud = "T";
00115     while (kitas_stud == "T" || kitas_stud == "t") {
00116         Student studentas;
00117         std::cout << "1 - įvesti pažymius rankiniu būdu\n2 - generuoti atsitiktinius pažymius\n3 -
generuoti vardus, pavardes ir pazymius\n4 - nuskaityti is failo\n5 - generuoti failus\n6 - baigti\n";
00118         std::cin >> chl;
00119         if (chl == 6) break;
00120         if (chl == 5)
00121         {
00122             std::cout<<"Ar nori generuoti failus? (T - taip, N - ne): ";
00123             std::string ats;
00124             std::cin >> ats;
00125             if (ats == "T" || ats == "t")
00126             {
00127                 std::vector<Student> vargsiukai;
00128                 std::vector<Student> kietekai;
00129                 std::vector<int> skaicius = {1000, 10000, 100000};
00130                 for (int i=0; i<skaicius.size(); i++)
00131                 {
00132                     std::string pav = "studentai_" + std::to_string(skaicius[i]) + ".txt";
00133                     gen_file(pav, skaicius[i]);
00134                 }
00135                 for (int i=0; i<skaicius.size(); i++)
00136                 {
00137                     std::string pav = "studentai_" + std::to_string(skaicius[i]) + ".txt";
00138                     failo_nuskaitymas(pav, skaicius[i], studentai);
00139                     std::cout<<"Failas studentai_"<<skaicius[i]<<"<.txt nuskaitytas"<<std::endl;
00140                 }
00141                 for (int i=0; i<skaicius.size(); i++)
00142                 {
00143                     std::string pav = "studentai_" + std::to_string(skaicius[i]) + ".txt";
00144                     padalinti_studentus(studentai, vargsiukai, kietekai);
00145                 }
00146                 issaugoti_studentus(vargsiukai, "vargsiukai.txt");
00147                 issaugoti_studentus(kietekai, "kietekai.txt");
00148             }
00149             if (ats == "N" || ats == "n")
00150             {
00151                 std::cout<<"Failai nesugeneruoti, naudojama esancius failus."<<std::endl;
00152             }
00153         }
00154     }
00155 }
00156 if (chl == 4) {
00157     try {
00158         std::ifstream rf("studentai10000.txt");
00159         if (!rf) throw std::runtime_error("Failas nerastas");
00160
00161         std::string var, pav, nd;
00162         rf >> var >> pav;
00163         int kiek = 0;
00164
00165         while (rf >> nd && nd != "Egz.") {
00166             kiek++;
00167         }

```

```

00168
00169         while (true) {
00170             std::string vardas, pavarde;
00171             std::vector<int> namuDarbai;
00172             int egz;
00173             if (!(rf » vardas » pavarde)) break;
00174
00175             for (int i = 0; i < kiek; i++) {
00176                 int paz;
00177                 if (!(rf » paz))
00178                     throw std::runtime_error("Klaida skaitant pažymius iš failo.");
00179                 namuDarbai.push_back(paz);
00180             }
00181
00182             if (!(egz))
00183                 throw std::runtime_error("Klaida skaitant egzamino pažymį iš failo.");
00184
00185             Student s;
00186             s.setVardas(vardas);
00187             s.setPavarde(pavarde);
00188             s.setNamuDarbai(namuDarbai);
00189             s.setEgz(egz);
00190
00191             studentai.push_back(s);
00192         }
00193     } catch (const std::exception& e) {
00194         std::cerr « e.what() « std::endl;
00195     }
00196 }
00197
00198 if (chl == 1) {
00199     std::string vardas, pavarde;
00200     std::vector<int> namuDarbai;
00201     int egz;
00202     try{
00203         std::cout « "Įveskite studento vardą: ";
00204         std::cin » vardas;
00205         if (std::isdigit(vardas[0])) throw std::runtime_error("Vardas negali prasidėti
skaičiumi");
00206         std::cout « "Įveskite studento pavardę: ";
00207         std::cin » pavarde;
00208         if (std::isdigit(pavarde[0])) throw std::runtime_error("Pavarde negali prasidėti
skaičiumi");
00209         std::cout « "Įveskite namų darbų pažymius (įveskite -1, kad baigtumėte): ";
00210         int pazymys;
00211         namuDarbai.clear();
00212         while (true) {
00213             std::cin » pazymys;
00214             if (pazymys < -1 || pazymys > 10) throw std::runtime_error("Pažymys turi būti
intervale nuo 1 iki 10");
00215             if (pazymys == -1) break;
00216             namuDarbai.push_back(pazymys);
00217         }
00218         if (std::cin.fail()) throw std::runtime_error("Neteisingas pažymys");
00219         std::cout « "Įveskite egzamino rezultata: ";
00220         std::cin » egz;
00221
00222         Student s;
00223         s.setVardas(vardas);
00224         s.setPavarde(pavarde);
00225         s.setNamuDarbai(namuDarbai);
00226         s.setEgz(egz);
00227
00228     } catch (const std::exception& e) {
00229         std::cerr « e.what() « std::endl;
00230     }
00231 }
00232
00233 if (chl == 2) {
00234     std::string vardas, pavarde;
00235     std::vector<int> namuDarbai;
00236     int egz;
00237     try{
00238         std::cout « "Įveskite studento vardą: ";
00239         std::cin » vardas;
00240         if (std::isdigit(vardas[0])) throw std::runtime_error("Vardas negali prasidėti
skaičiumi");
00241         std::cout « "Įveskite studento pavardę: ";
00242         std::cin » pavarde;
00243         if (std::isdigit(pavarde[0])) throw std::runtime_error("Pavarde negali prasidėti
skaičiumi");
00244         namuDarbai.clear();
00245         for (int i = 0; i < 5; i++) {
00246             namuDarbai.push_back(distr(gen));
00247         }
00248         egz = distr(gen);
00249     } catch (const std::exception& e) {

```

```

00250         std::cerr << e.what() << std::endl;
00251     }
00252     Student s;
00253     s.setVardas(vardas);
00254     s.setPavarde(pavarde);
00255     s.setNamuDarbai(namuDarbai);
00256     s.setEgz(egz);
00257 }
00258
00259 if (chl == 3) {
00260     std::string vardas, pavarde;
00261     std::vector<int> namuDarbai;
00262     int egz;
00263
00264     gen_name(vardas, pavarde);
00265     namuDarbai.clear();
00266     for (int i = 0; i < 5; i++) {
00267         namuDarbai.push_back(distr(gen));
00268     }
00269     egz = distr(gen);
00270
00271     Student s;
00272     s.setVardas(vardas);
00273     s.setPavarde(pavarde);
00274     s.setNamuDarbai(namuDarbai);
00275     s.setEgz(egz);
00276 }
00277
00278     std::cout << "Ar norite tēsti? (T - taip, N - ne): ";
00279     std::cin >> kitas_stud;
00281 }
00282 }
00283
00284 void Rez(std::vector<Student>& studentai, int& chl) {
00285     while (chl != 5) {
00286         std::cout << "Pasirinkite rikiavimo būdą:\n1 - pagal vardą\n2 - pagal pavardę\n3 - pagal
vidurkį\n4 - pagal medianą\n";
00287         int ch;
00288         std::cin >> ch;
00289         switch (ch) {
00290             case 1:
00291                 std::sort(studentai.begin(), studentai.end(), [](const Student& a, const Student& b) {
00292                     return a.getVardas() < b.getVardas();
00293                 });
00294                 break;
00295             case 2:
00296                 std::sort(studentai.begin(), studentai.end(), [](const Student& a, const Student& b) {
00297                     return a.getPavarde() < b.getPavarde();
00298                 });
00299                 break;
00300             case 3:
00301                 std::sort(studentai.begin(), studentai.end(), [](const Student& a, const Student& b) {
00302                     return a.getGalutinisVid() < b.getGalutinisVid();
00303                 });
00304                 break;
00305             case 4:
00306                 std::sort(studentai.begin(), studentai.end(), [](const Student& a, const Student& b) {
00307                     return a.getGalutinisMed() < b.getGalutinisMed();
00308                 });
00309                 break;
00310             default:
00311                 std::cout << "Neteisingas pasirinkimas\n";
00312                 return;
00313         }
00314         std::cout << std::fixed << std::setprecision(2);
00315         std::cout << "-----\n";
00316         std::cout << std::setw(15) << "Vardas" << std::setw(15) << "Pavarde" << std::setw(25) << "Galutinis
(vid.) Galutinis (med.)\n";
00317         std::cout << "-----\n";
00318
00319         for (const auto& s : studentai) {
00320             std::cout << std::setw(15) << s.getVardas()
00321                 << std::setw(15) << s.getPavarde()
00322                 << std::setw(15) << s.getGalutinisVid()
00323                 << std::setw(15) << s.getGalutinisMed() << "\n";
00324         }
00325         std::cout << "Ar norite išsaugoti į failą? (T - taip, N - ne): ";
00326         std::string ats;
00327         std::cin >> ats;
00328         if (ats == "T" || ats == "t") {
00329             std::ofstream rf("rezultatai.txt");
00330
00331             rf << std::fixed << std::setprecision(2);
00332             rf << "-----\n";
00333             rf << std::setw(15) << "Vardas" << std::setw(15) << "Pavarde" << std::setw(25) << "Galutinis (vid.)
Galutinis (med.)\n";

```

```

00334         rf « "-----\n";
00335         for (const auto& s : studentai) {
00336
00337             rf « std::setw(15) « s.getVardas()
00338                « std::setw(15) « s.getPavarde()
00339                « std::setw(15) « s.getGalutinisVid()
00340                « std::setw(15) « s.getGalutinisMed() « "\n";
00341         }
00342     }
00343 }
00344
00345 }

```

6.6 functions.h File Reference

```

#include "student.h"
#include "functions.cpp"
#include <vector>
#include <string>

```

Functions

- double [Vidurkis](#) (const std::vector< int > &pazymiai)
- double [apskaiciuotiMediana](#) (std::vector< int > pazymiai)
- void [gen_name](#) (std::string &vardas, std::string &pavarde)
- void [Duom](#) (std::vector< [Student](#) > &studentai)
- void [Rez](#) (std::vector< [Student](#) > &studentai)

6.6.1 Function Documentation

6.6.1.1 [apskaiciuotiMediana\(\)](#)

```

double apskaiciuotiMediana (
    std::vector< int > pazymiai)

```

6.6.1.2 [Duom\(\)](#)

```

void Duom (
    std::vector< Student > & studentai)

```

6.6.1.3 [gen_name\(\)](#)

```

void gen_name (
    std::string & vardas,
    std::string & pavarde)

```

6.6.1.4 [Rez\(\)](#)

```

void Rez (
    std::vector< Student > & studentai)

```

6.6.1.5 Vidurkis()

```
double Vidurkis (
    const std::vector< int > & pazymiai)
```

6.7 functions.h

[Go to the documentation of this file.](#)

```
00001 #ifndef FUNCTIONS_H
00002 #define FUNCTIONS_H
00003
00004 #include "student.h"
00005 #include "functions.cpp"
00006 #include <vector>
00007 #include <string>
00008
00009 double Vidurkis(const std::vector<int>& pazymiai);
00010 double apskaiciuotiMediana(std::vector<int> pazymiai);
00011 void gen_name(std::string& vardas, std::string& pavarde);
00012 void Duom(std::vector<Student>& studentai);
00013 void Rez(std::vector<Student>& studentai);
00014
00015 #endif
```

6.8 kont_test.cpp File Reference

```
#include <iostream>
#include <vector>
#include <list>
#include <deque>
#include <algorithm>
#include <chrono>
#include <random>
```

Classes

- class [Student](#)

Functions

- template<typename Container>
void [generate_students](#) (Container &students, size_t count)
- void [sort_students](#) (std::list< [Student](#) > &students)
- template<typename Container>
void [sort_students](#) (Container &students)
- template<typename Container>
void [benchmark](#) (size_t count)
- int [main](#) ()

6.8.1 Function Documentation

6.8.1.1 benchmark()

```
template<typename Container>
void benchmark (
    size_t count)
```

6.8.1.2 generate_students()

```
template<typename Container>
void generate_students (
    Container & students,
    size_t count)
```

6.8.1.3 main()

```
int main ()
```

6.8.1.4 sort_students() [1/2]

```
template<typename Container>
void sort_students (
    Container & students)
```

6.8.1.5 sort_students() [2/2]

```
void sort_students (
    std::list< Student > & students)
```

6.9 list_test.cpp File Reference

```
#include <iostream>
#include <fstream>
#include <vector>
#include <string>
#include <algorithm>
#include <numeric>
#include <iomanip>
#include <chrono>
#include <list>
```

Classes

- class [Student](#)

Functions

- void `failo_nuskaitymas` (const std::string &pav, std::list< `Student` > &studentai)
- void `padalinti1` (const std::list< `Student` > &studentai, std::list< `Student` > &kietekai, std::list< `Student` > &vargsiukai)
- void `padalinti2` (std::list< `Student` > &studentai, std::list< `Student` > &vargsiukai)
- void `padalinti3` (std::list< `Student` > &studentai, std::list< `Student` > &vargsiukai)
- void `issaugoti_studentus` (const std::list< `Student` > &studentai, const std::string &failo_pav)
- int `main` ()

6.9.1 Function Documentation

6.9.1.1 failo_nuskaitymas()

```
void failo_nuskaitymas (  
    const std::string & pav,  
    std::list< Student > & studentai)
```

6.9.1.2 issaugoti_studentus()

```
void issaugoti_studentus (  
    const std::list< Student > & studentai,  
    const std::string & failo_pav)
```

6.9.1.3 main()

```
int main ()
```

6.9.1.4 padalinti1()

```
void padalinti1 (  
    const std::list< Student > & studentai,  
    std::list< Student > & kietekai,  
    std::list< Student > & vargsiukai)
```

6.9.1.5 padalinti2()

```
void padalinti2 (  
    std::list< Student > & studentai,  
    std::list< Student > & vargsiukai)
```

6.9.1.6 padalinti3()

```
void padalinti3 (  
    std::list< Student > & studentai,  
    std::list< Student > & vargsiukai)
```

6.10 main.cpp File Reference

```
#include "functions.h"
#include <iostream>
#include <vector>
#include <string>
#include <iomanip>
#include <algorithm>
```

Functions

- double [Vidurkis](#) (const std::vector< int > &pazymiai)
- double [apskaiciuotiMediana](#) (std::vector< int > pazymiai)
- void [Duom](#) ([Student](#) &studentas, std::vector< [Student](#) > &studentai)
- void [Rez](#) (const std::vector< [Student](#) > &studentai, bool n_vid)
- int [main](#) ()

6.10.1 Function Documentation

6.10.1.1 [apskaiciuotiMediana\(\)](#)

```
double apskaiciuotiMediana (
    std::vector< int > pazymiai)
```

6.10.1.2 [Duom\(\)](#)

```
void Duom (
    Student & studentas,
    std::vector< Student > & studentai)
```

6.10.1.3 [main\(\)](#)

```
int main ()
```

6.10.1.4 [Rez\(\)](#)

```
void Rez (
    const std::vector< Student > & studentai,
    bool n_vid)
```

6.10.1.5 [Vidurkis\(\)](#)

```
double Vidurkis (
    const std::vector< int > & pazymiai)
```

6.11 masyvas.cpp File Reference

```
#include "func_masyvas.h"
```

Functions

- double [Vidurkis](#) (const [Student](#) &studentas)
- double [apskaiciuotiMediana](#) (const [Student](#) &studentas)
- void [gen_name](#) (std::string &vardas, std::string &pavarde)
- void [Duom](#) ([Student](#) &studentas, std::vector< [Student](#) > &studentai)
- void [Rez](#) (const [Student](#) &studentas, bool n_vid)
- int [main](#) ()

6.11.1 Function Documentation

6.11.1.1 [apskaiciuotiMediana\(\)](#)

```
double apskaiciuotiMediana (  
    const Student & studentas)
```

6.11.1.2 [Duom\(\)](#)

```
void Duom (  
    Student & studentas,  
    std::vector< Student > & studentai)
```

6.11.1.3 [gen_name\(\)](#)

```
void gen_name (  
    std::string & vardas,  
    std::string & pavarde)
```

6.11.1.4 [main\(\)](#)

```
int main ()
```

6.11.1.5 [Rez\(\)](#)

```
void Rez (  
    const Student & studentas,  
    bool n_vid)
```

6.11.1.6 Vidurkis()

```
double Vidurkis (  
    const Student & studentas)
```

6.12 README.md File Reference

6.13 ro5_test.cpp File Reference

```
#include "Student.h"  
#include "Zmogus.h"  
#include <sstream>  
#include <iostream>
```

Functions

- `std::istream & operator>>` (`std::istream &is`, `Student &s`)
- `std::ostream & operator<<` (`std::ostream &os`, `const Student &s`)
- `void base_test` (`Zmogus &z`)
- `void ro5_test` ()
- `int main` ()

6.13.1 Function Documentation

6.13.1.1 base_test()

```
void base_test (  
    Zmogus & z)
```

6.13.1.2 main()

```
int main ()
```

6.13.1.3 operator<<()

```
std::ostream & operator<< (  
    std::ostream & os,  
    const Student & s)
```

6.13.1.4 operator>>()

```
std::istream & operator>> (  
    std::istream & is,  
    Student & s)
```

6.13.1.5 ro5_test()

```
void ro5_test ()
```

6.14 student.h File Reference

```
#include <iostream>
#include <vector>
#include <string>
#include <algorithm>
#include "zmogus.h"
```

Classes

- class [Student](#)

6.15 student.h

[Go to the documentation of this file.](#)

```
00001 #ifndef STUDENT_H
00002 #define STUDENT_H
00003
00004 #include <iostream>
00005 #include <vector>
00006 #include <string>
00007 #include <algorithm>
00008 #include "zmogus.h"
00009
00010 class Student : public Zmogus{
00011 private:
00012
00013     std::vector<int> namuDarbai;
00014     int egz;
00015 public:
00016     Student() : Zmogus("", ""), egz(0) {}
00017
00018     Student(const std::string& v, const std::string& p, const std::vector<int>& nd, int e)
00019     : Zmogus(v, p), namuDarbai(nd), egz(e) {}
00020
00021
00022     Student(const Student& old_stud)
00023     : Zmogus(old_stud.vardas, old_stud.pavarde),
00024       namuDarbai(old_stud.namuDarbai), egz(old_stud.egz) {}
00025
00026
00027
00028     Student(Student&& other) noexcept
00029     : Zmogus(std::move(other.vardas), std::move(other.pavarde)),
00030       namuDarbai(std::move(other.namuDarbai)),
00031       egz(other.egz) {}
00032
00033     Student& operator=(const Student& other) {
00034     if (this != &other) {
00035         setVardas(other.getVardas());
00036         setPavarde(other.getPavarde());
00037         namuDarbai = other.namuDarbai;
00038         egz = other.egz;
00039     }
00040     return *this;
00041 }
00042
00043
00044     Student& operator=(Student&& other) noexcept {
00045     if (this != &other) {
00046         setVardas(std::move(other.getVardas()));
00047         setPavarde(std::move(other.getPavarde()));
```

```

00048         namuDarbai = std::move(other.namuDarbai);
00049         egz = other.egz;
00050     }
00051     return *this;
00052 }
00053
00054 ~Student() = default;
00055
00056 void setVardas(const std::string& v) { vardas = v; }
00057 void setPavarde(const std::string& p) { pavarde = p; }
00058 void setNamuDarbai(const std::vector<int>& nd) { namuDarbai = nd; }
00059 void setEgz(int e) { egz = e; }
00060
00061 std::string getVardas() const { return vardas; }
00062 std::string getPavarde() const { return pavarde; }
00063 std::vector<int> getNamuDarbai() const { return namuDarbai; }
00064 int getEgz() const { return egz; }
00065 double getGalutinisVid() const { return skaiciuotiGalutiniVidurki(); }
00066 double getGalutinisMed() const { return skaiciuotiGalutiniMediana(); }
00067
00068 static double Vidurkis(const std::vector<int>& pazymiai) {
00069     if (pazymiai.empty()) return 0.0;
00070     double suma = 0.0;
00071     for (int pazymys : pazymiai) suma += pazymys;
00072     return suma / pazymiai.size();
00073 }
00074
00075 double skaiciuotiVidurki() const {
00076     if (namuDarbai.empty()) return 0.0;
00077     double suma = 0.0;
00078     for (int pazymys : namuDarbai) {
00079         suma += pazymys;
00080     }
00081     return suma / namuDarbai.size();
00082 }
00083
00084 double skaiciuotiMediana() const {
00085     if (namuDarbai.empty()) return 0.0;
00086     std::vector<int> temp = namuDarbai;
00087     std::sort(temp.begin(), temp.end());
00088     size_t dydis = temp.size();
00089     if (dydis % 2 == 0) {
00090         return (temp[dydis / 2 - 1] + temp[dydis / 2]) / 2.0;
00091     } else {
00092         return temp[dydis / 2];
00093     }
00094 }
00095
00096 double skaiciuotiGalutiniVidurki() const {
00097     return skaiciuotiVidurki() * 0.4 + egz * 0.6;
00098 }
00099
00100 double skaiciuotiGalutiniMediana() const {
00101     return skaiciuotiMediana() * 0.4 + egz * 0.6;
00102 }
00103
00104 friend std::ostream& operator<<(std::ostream& os, const Student& s);
00105 friend std::istream& operator>>(std::istream& is, Student& s);
00106 };
00107 #endif

```

6.16 vector_test.cpp File Reference

```

#include <iostream>
#include <fstream>
#include <vector>
#include <string>
#include <algorithm>
#include <numeric>
#include <iomanip>
#include <chrono>

```

Classes

- class [Student](#)

Functions

- void `failo_nuskaitymas` (const std::string &pav, std::vector< `Student` > &studentai)
- void `padalinti1` (const std::vector< `Student` > &studentai, std::vector< `Student` > &kietekai, std::vector< `Student` > &vargsiukai)
- void `padalinti2` (std::vector< `Student` > &studentai, std::vector< `Student` > &vargsiukai)
- void `padalinti3` (std::vector< `Student` > &studentai, std::vector< `Student` > &vargsiukai)
- void `issaugoti_studentus` (const std::vector< `Student` > &studentai, const std::string &failo_pav)
- int `main` ()

6.16.1 Function Documentation

6.16.1.1 failo_nuskaitymas()

```
void failo_nuskaitymas (  
    const std::string & pav,  
    std::vector< Student > & studentai)
```

6.16.1.2 issaugoti_studentus()

```
void issaugoti_studentus (  
    const std::vector< Student > & studentai,  
    const std::string & failo_pav)
```

6.16.1.3 main()

```
int main ()
```

6.16.1.4 padalinti1()

```
void padalinti1 (  
    const std::vector< Student > & studentai,  
    std::vector< Student > & kietekai,  
    std::vector< Student > & vargsiukai)
```

6.16.1.5 padalinti2()

```
void padalinti2 (  
    std::vector< Student > & studentai,  
    std::vector< Student > & vargsiukai)
```

6.16.1.6 padalinti3()

```
void padalinti3 (  
    std::vector< Student > & studentai,  
    std::vector< Student > & vargsiukai)
```

6.17 vektoriai.cpp File Reference

```
#include "functions.h"
#include "student.h"
#include <chrono>
#include <iostream>
```

Functions

- int [main](#) ()

6.17.1 Function Documentation

6.17.1.1 main()

```
int main ()
```

6.18 zmogus.h File Reference

```
#include <string>
```

Classes

- class [Zmogus](#)

6.19 zmogus.h

[Go to the documentation of this file.](#)

```
00001 #ifndef ZMOGUS_H
00002 #define ZMOGUS_H
00003
00004 #include <string>
00005
00006 class Zmogus {
00007 protected:
00008     std::string vardas;
00009     std::string pavarde;
00010
00011 public:
00012     Zmogus() = default;
00013     Zmogus(const std::string& v, const std::string& p) : vardas(v), pavarde(p) {}
00014     virtual ~Zmogus() = default;
00015
00016     virtual std::string getVardas() const = 0;
00017     virtual std::string getPavarde() const = 0;
00018
00019     virtual void setVardas(const std::string& v) = 0;
00020     virtual void setPavarde(const std::string& p) = 0;
00021 };
00022
00023 #endif
```

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